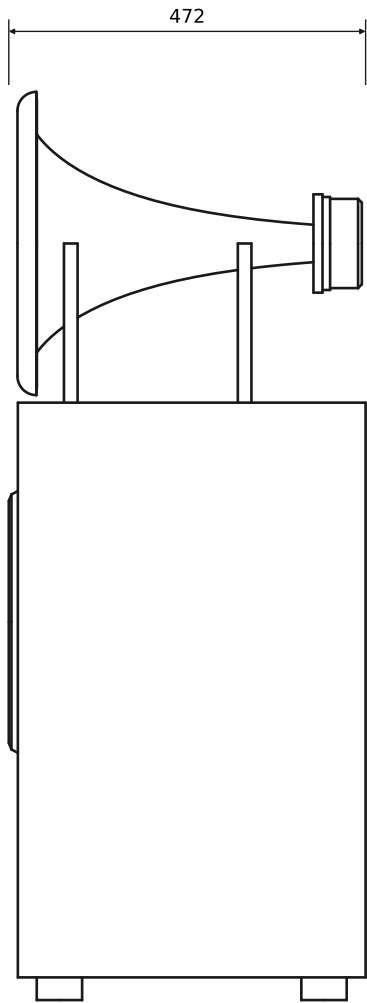
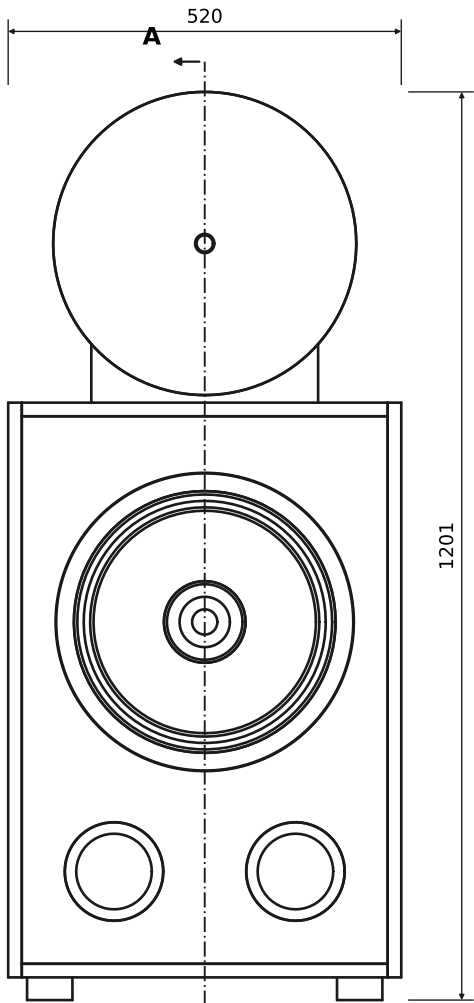


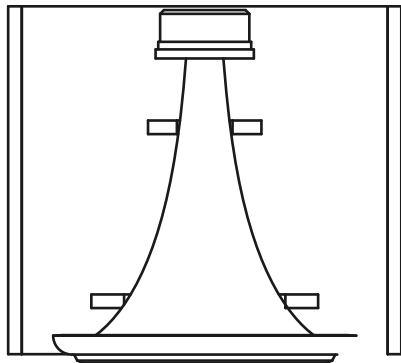
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



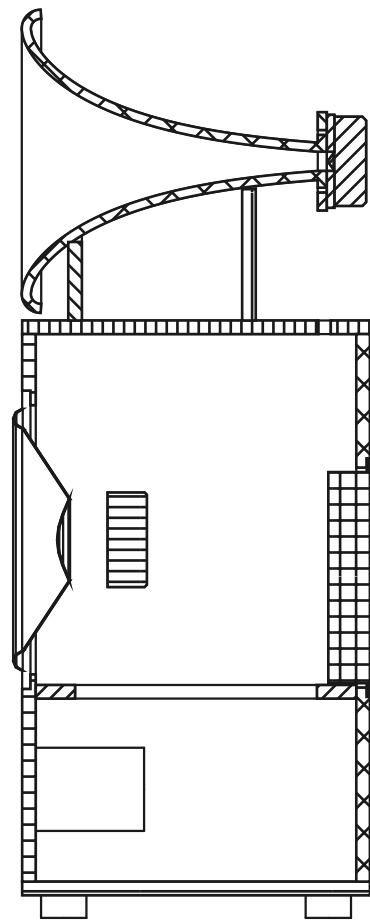
FROM ABOVE

General arrangement

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 260.0 and seen from the right; cut faces hatched, a different angle for each part.

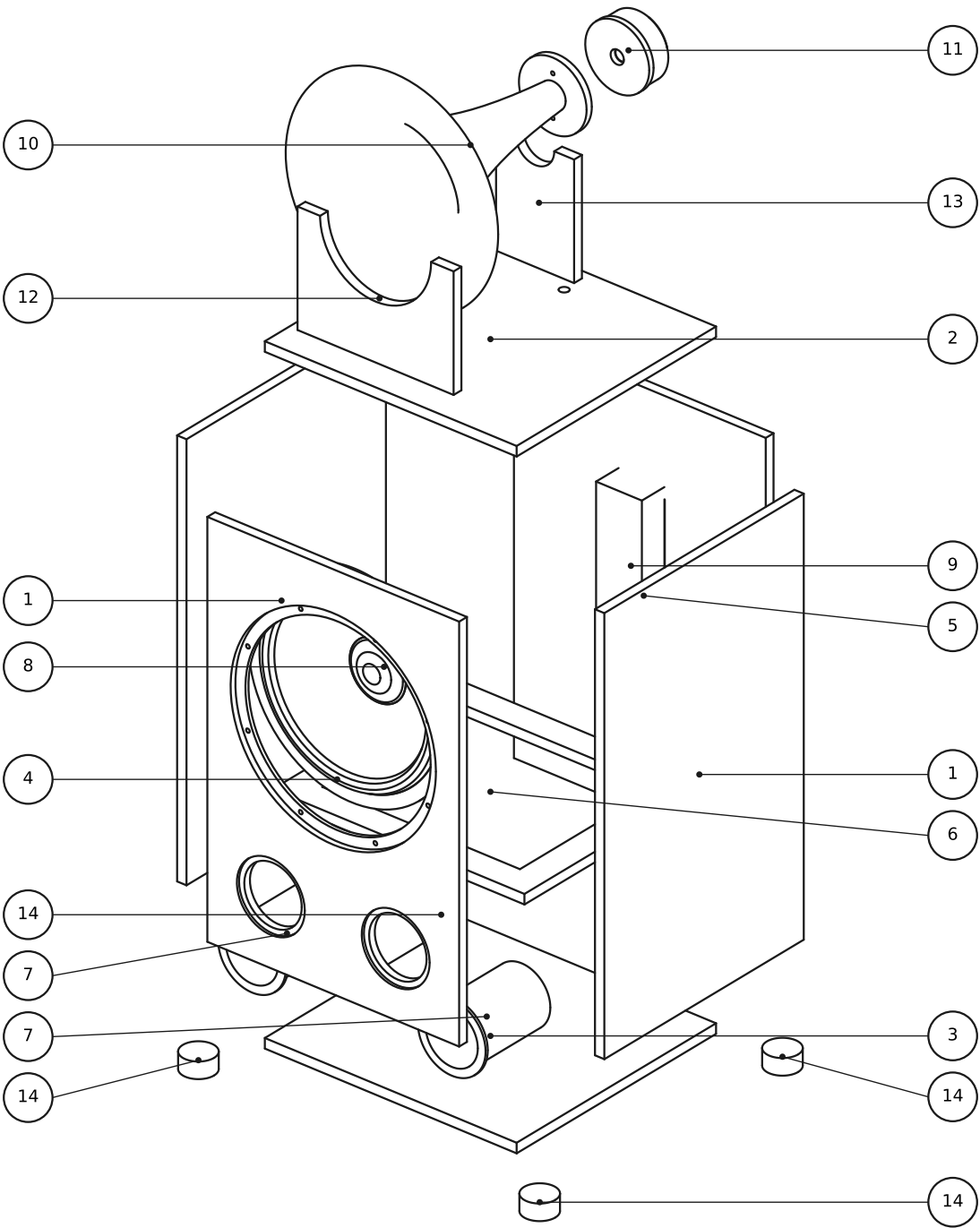


Section A-A

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).

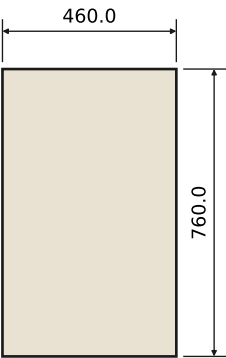


Exploded view

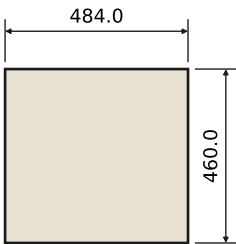
barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

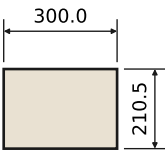
Scale 1:20. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



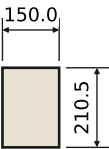
left, right
18 mm baltic-birch, 4 for the set



bottom
18 mm baltic-birch, 2 for the set



saddle-1
18 mm baltic-birch, 2 for the set



saddle-2
18 mm baltic-birch, 2 for the set

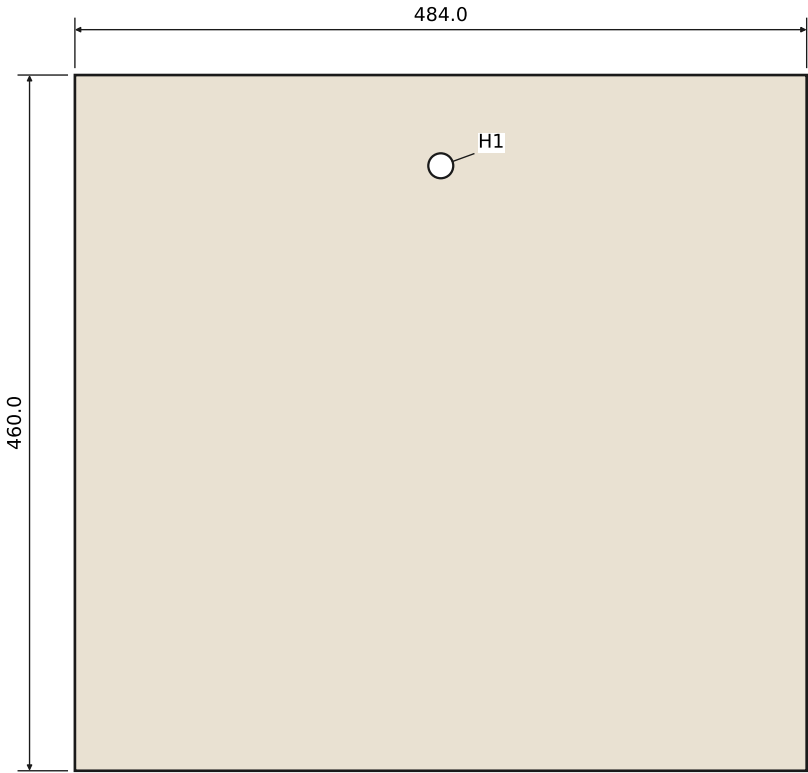
Sheet parts: plain panels

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø16.5	242.0	400.0	through
H: cut through. P: pocket from the face shown. U: pocket from the inner face (dashed; its own DXF). Centres from the lower-left corner, seen from the outer face.				



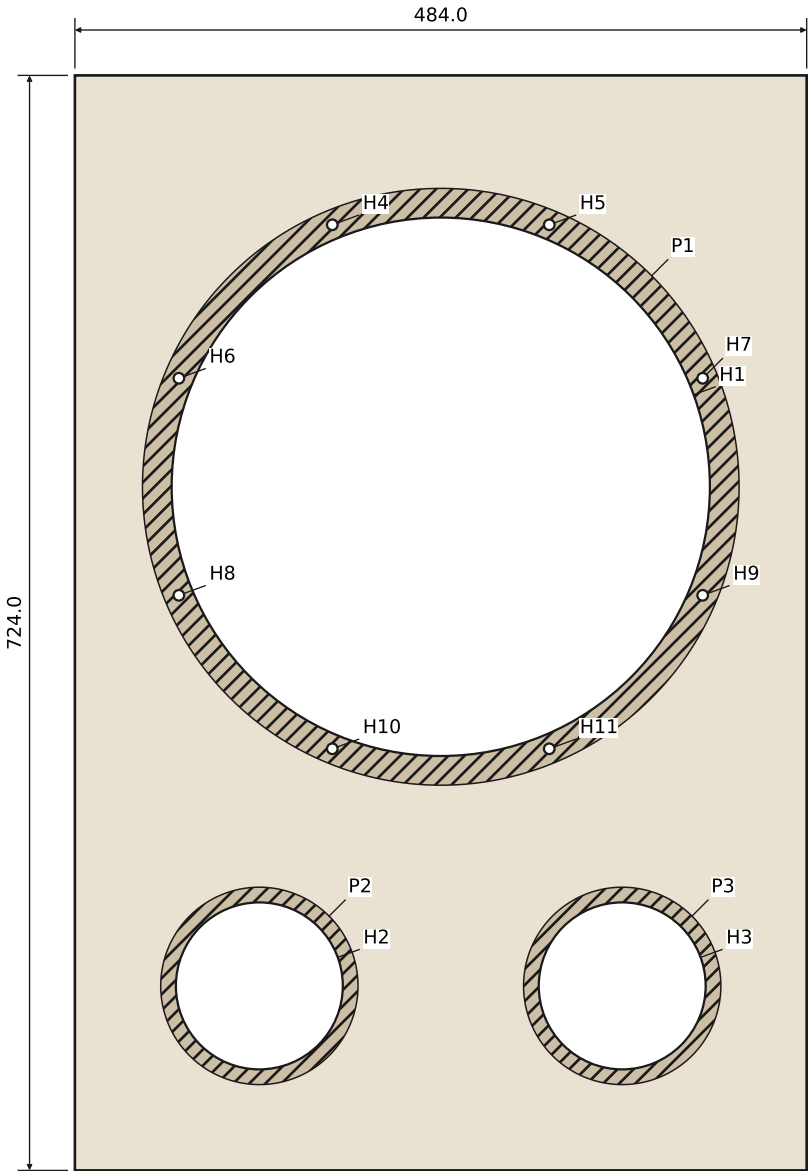
top: 18 mm baltic-birch, 2 for the set, seen from its outer face

Sheet part: top

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/baffle.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø394.6	242.0	452.0	11 from face
P2	Ø130.5	122.0	122.0	5 from face
P3	Ø130.5	362.0	122.0	5 from face
H1	Ø356.0	242.0	452.0	through
H2	Ø110.4	122.0	122.0	through
H3	Ø110.4	362.0	122.0	through
H4	Ø7.0	170.2	625.2	through
H5	Ø7.0	313.8	625.2	through
H6	Ø7.0	68.8	523.8	through
H7	Ø7.0	415.2	523.8	through
H8	Ø7.0	68.8	380.2	through
H9	Ø7.0	415.2	380.2	through
H10	Ø7.0	170.2	278.8	through
H11	Ø7.0	313.8	278.8	through

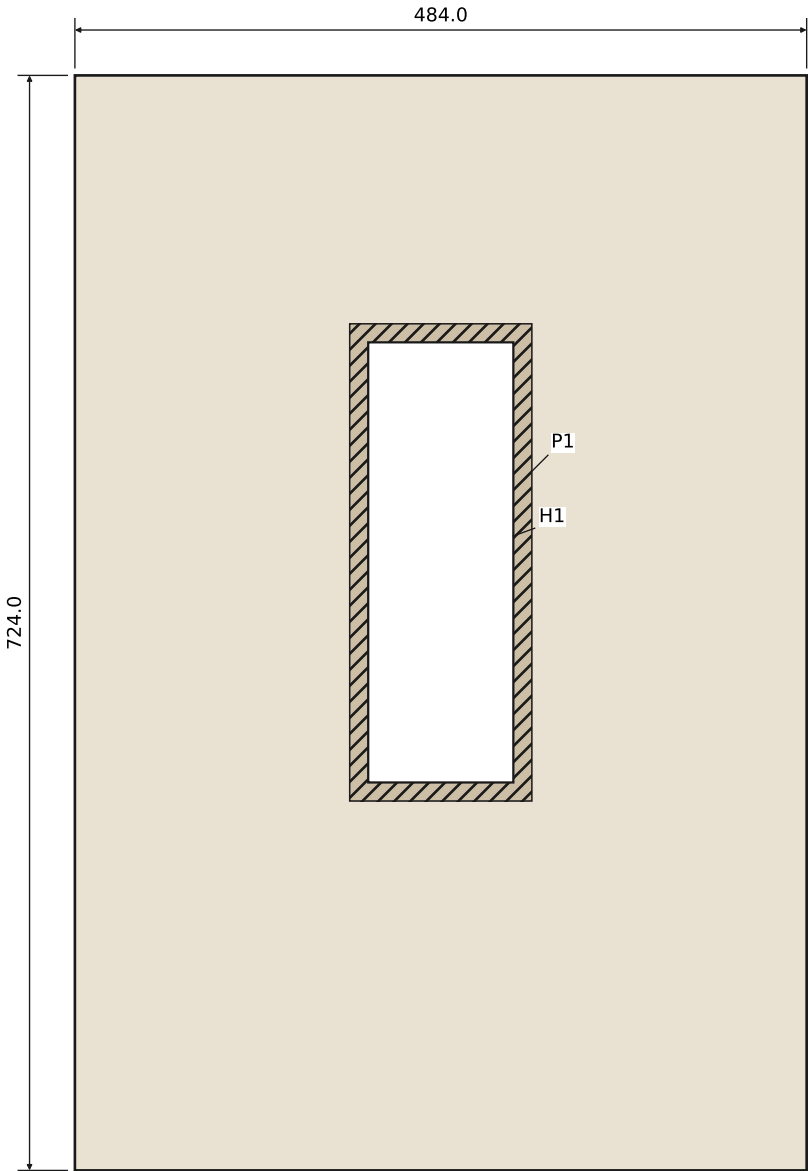
H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

baffle: 18 mm baltic-birch, 2 for the set, seen from its outer face

Sheet part: baffle

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

Scale 1:5. As the router sees it from the outer face; DXF:
dxf/back.dxf. Marks listed at the right.



back: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	242.0	402.0	4.5 from face
H1	96.0 x 291.0	242.0	402.0	through

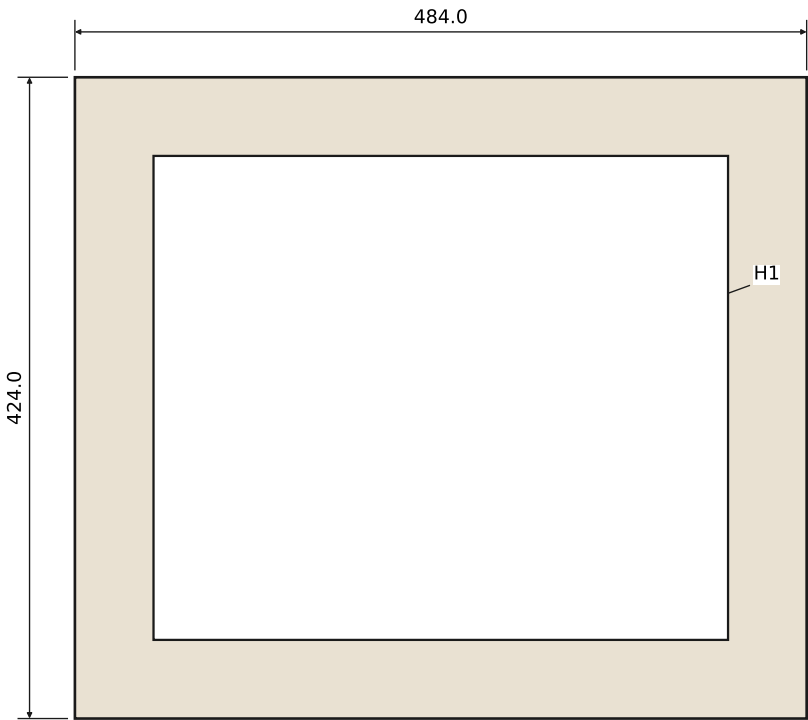
H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	380.0 x 320.0	242.0	212.0	through
H: cut through. P: pocket from the face shown.				
U: pocket from the inner face (dashed; its own DXF).				
Centres from the lower-left corner, seen from the outer face.				



brace: 18 mm baltic-birch, 2 for the set, seen from its outer face

Sheet part: brace

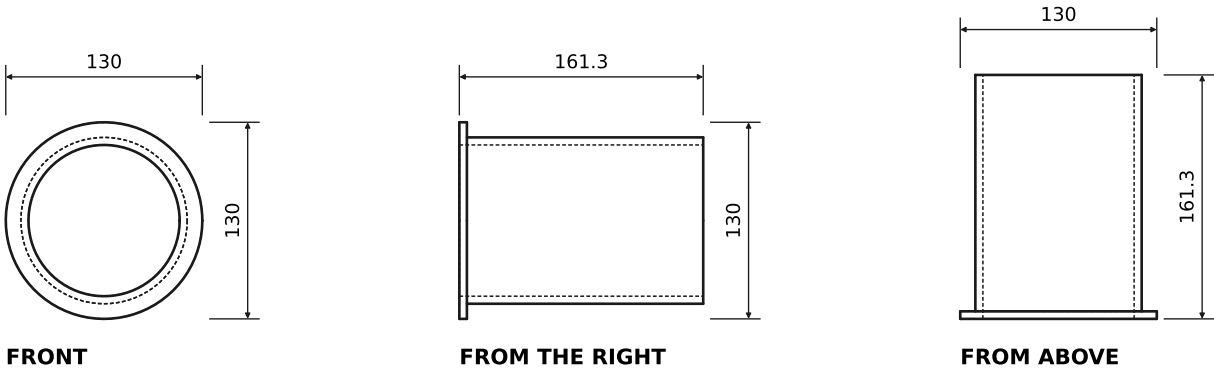
barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

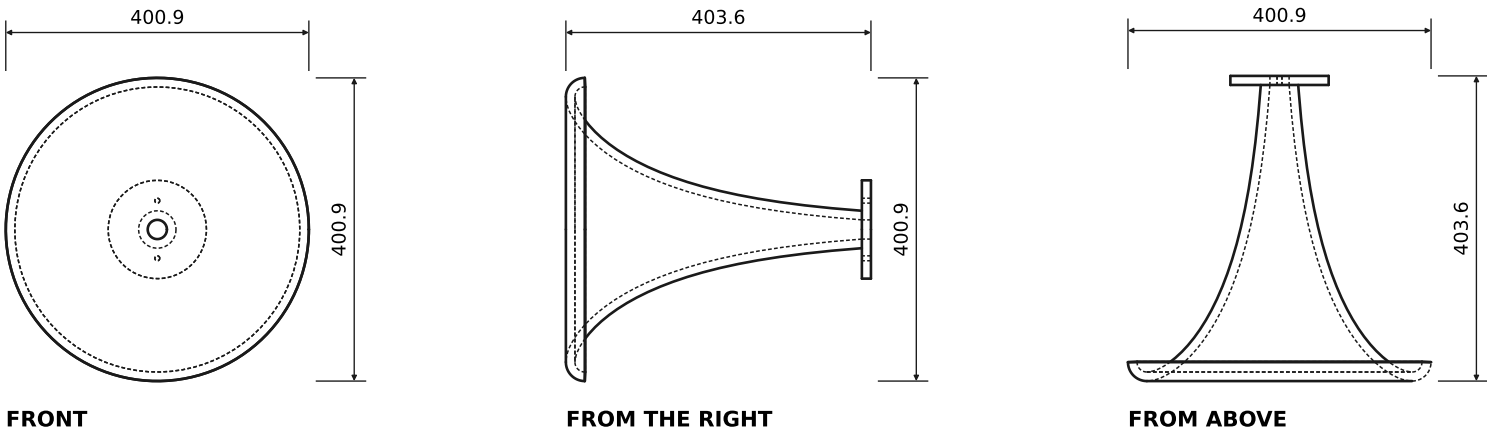
7. port-1, port-2: asa print, flange down; 285 cm3 each; 4 for the set; 1:5

161 long tunes the box to 38 Hz; print 15 longer and trim to tune by impedance



10. horn: cnc-lathe, birch-hardwood; 2912 cm3 each; 2 for the set; 1:10

tractrix to 320 Hz, 1 in throat, mouth 341 across, 391 long, 12 wall, lip rolled on 25: laminated birch rings, turned, sealed and painted (or the boug



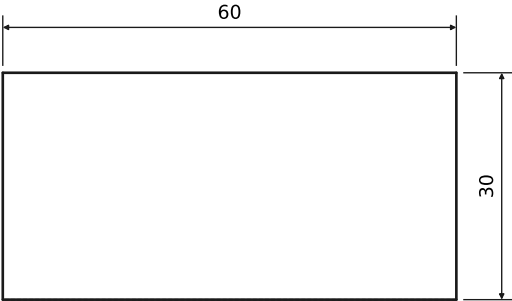
Printed and machined parts

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

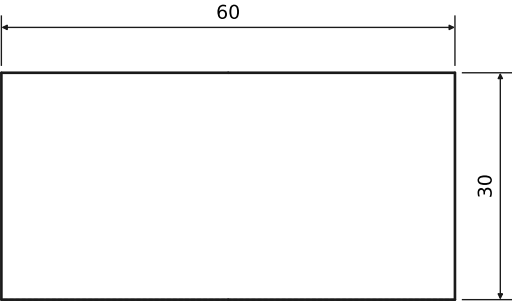
mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

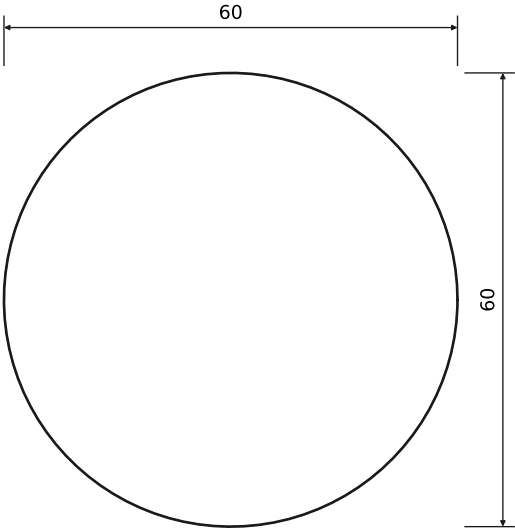
14. foot-1, foot-2, foot-3, foot-4: lathe, oak; 85 cm3 each; 8 for the set; 1:1



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark

mm | issue 2026-10-10 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	18 mm Baltic birch plywood, BB/BB	460 x 760	25.35
2	top	cut	2	18 mm Baltic birch plywood, BB/BB	484 x 460	16.14
3	bottom	cut	2	18 mm Baltic birch plywood, BB/BB	484 x 460	16.14
4	baffle	cut	2	18 mm Baltic birch plywood, BB/BB	484 x 724	25.41
5	back	cut	2	18 mm Baltic birch plywood, BB/BB	484 x 724	25.41
6	brace	cut	2	18 mm Baltic birch plywood, BB/BB	484 x 424	14.88
7	port-1, port-2	print	4	ASA, FDM (outdoor, paintable)	285 cm3	27.94
8	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
9	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
10	horn	cnc-lathe	2	Birch, solid	411 x 414 x 411	862.99
11	hf	buy	2	BMS 4550-16 compression driver, 1 in exit, polyester diaphragm, 16 ohm	123 across, 52 deep, 2 x M6 on 76.2	154.00
12	saddle-1	cut	2	18 mm Baltic birch plywood, BB/BB	300 x 210	4.58
13	saddle-2	cut	2	18 mm Baltic birch plywood, BB/BB	150 x 210	2.29
14	foot-1, foot-2, foot-3, foot-4	lathe	8	White oak, solid	70 x 70 x 40	13.53
15	woofer-fixings	buy	4	M4 x 20 low-profile button heads (A2) with M4 T-nuts, a driver's set of 4	M4	4.00
16	hf-fixings	buy	2	M6 x 25 socket caps with washers (the compression driver to the horn's flange), a pair		[PRICE]
17	gasket	buy	4	1.5 mm closed-cell foam gasket strip, 1 m		5.00
18	damping	buy	4	Polyester fibre fill, 150 g		8.00
19	amp-fixings	buy	2	Hypex 4.3 x 16 self-tapping pan heads and 3 mm EPDM tape, a plate's set		10.00
20	cable	buy	6	Round-sheathed speaker cable 2 x 1.5 mm2, 1 m		2.00
21	terminals	buy	8	6.3 mm insulated push-on terminal		0.50
22	hf-pad	buy	2	Resistor L-pad for the HF channel, about -10 dB at 16 ohm (lowers the amplifier's hiss through a 113 dB driver)		[PRICE]
23	strap	buy	2	Leather strap 25 mm with a brass buckle, to hold the horn in its saddles		[PRICE]
24	felt	buy	2	Wool felt tape 3 mm, self-adhesive, 1 m (lines the saddles)		[PRICE]

Notes

1. Bark: real birch bark sheet (Betula papyrifera, from fallen or felled trees) laminated to the plywood with PVA in a vacuum bag, sealed with a matt waterborne lacquer; for eucalyptus, the shed bark of a smooth-barked gum (spotted or snow gum) pressed flat and laminated the same way, or a UV-printed veneer of it. Approve a sample board first.
2. Glue-up: sides to the bottom, the brace, then the baffle and back, then the top.
3. The woofer flush in its rebate on a gasket, eight M6 T-nuts from inside; the ports pushed in dry, trimmed to tune (impedance minimum at fb), then bonded.
4. The FA122 crosses at about 800 Hz (LR4) with the HF channel about 14 dB down, part of it in the passive L-pad after the amplifier, so its hiss stays inaudible through a 113 dB driver.
5. The horn rests in its felt-lined saddles, strapped down; its cable runs through an M16 gland in the top.
6. Hypex advises a sealed box round the amplifier module (left out here, as in the kit's example).

Parts list and notes

barkhorn: Two-way hybrid horn loudspeaker, birch bark or eucalyptus bark